Solving the two equations gives

## IV.

Reference solution:

For convenience, all heats below denote magnitudes.

Combine an engine and heat pump: the engine converts boiler heat to work, which drives the pump to absorb outdoor heat and deliver it indoors.

The boiler is the engine's hot source; a cold source can be indoors or outdoors. For the room, let the engine absorb from the boiler and reject indoors. Then

The pump's hot source is indoors, its cold source outdoors. It absorbs outdoors and rejects indoors. Then

The room receives a total heat from the combined cycle of

Adding gives

The same result follows if the engine uses outdoors as its cold source.

Energy conservation gives

To deliver directly to the room, the boiler must release

Combining gives the boiler heat for combined heating:

The fractional cost reduction is